Session 3: Let the loop do the walking

Question: Can one rule label every order line?

Concepts

Concept	In daily life	In Excel	The limit
for loop	Walking down the shelf, checking each bin	Dragging a formula down	The body must be indented
function	Your own reusable rule	Your own named formula	It does nothing until you call it
return	The answer the rule hands back	The value the formula shows	print only shows. return hands it back
traceback	A stack of plates, newest on top	None	Read the bottom line first, then go up

Code patterns

```
for quantity in quantities:
    print(quantity)

def order_size(quantity):
    if quantity >= 100:
        return "big"
    elif quantity >= 12:
        return "medium"
    else:
        return "small"

for quantity in quantities:
    print(f"{quantity} -> {order_size(quantity)}")
```

Three common errors

Error	What to do
IndentationError	The lines inside the loop or function are not indented.
None instead of an answer	The function uses print instead of return.
NameError for a function	You called the function before you ran the cell with `def`.

Exercises and expected answers

Exercise	Expected
Print each quantity times 2	12, 12, 16, 12, 12, 4, 12
line_value(6, 2.55)	15.3
Values of the 5 real lines	15.3, 20.34, 22.0, 20.34, 20.34
Label 6, 12, 99, 100, 1930	small, medium, medium, big, big